

Nature | Volume 656 | Issue 8128

Issue date: 2026-08-20 | Retrieved: 2026-08-20T01:54:53.078732+00:00 | 14 relevant primary-research papers

Evidence discipline

Paper-supported findings, published enabling evidence, cross-species inference, speculative hypotheses, and proposed pilot experiments are labeled separately. Access notes state when only an abstract or limited preview was available.

Screening notes

- Official publisher issue page was reachable and contained the detected volume and issue identifiers.
- Screened 14 biology-relevant journal articles after excluding reviews, perspectives, news, corrections, and editorials by metadata/title rules.
- All summaries are deterministic extractions from the bounded public evidence packet; no language-model API or paid research service was used.

Official issue

<https://www.nature.com/nature/volumes/656/issues/8128>

Why Xenopus now

Early Xenopus combines rapid external development, targeted microinjection, explants, lineage tracing, live imaging, F0 CRISPR, rescue, and scalable replication. It can convert abstract-supported candidate mechanisms into spatially resolved causal tests before committing to slower mammalian models.

Whole-genome duplication shaped cell-type evolution in the vertebrate brain

Yuanzhen Zhu, Shuai Zhang, Jiankai Wei, Diego Dolgetta-Garcia, Katia Jindrich, Huimin Liu, Chenggang Shi, Rongrong Pan, et al.

journal-article | Nature 656(8128) | 2026-08-20

DOI 10.1038/s41586-026-10629-x

Access: Publisher-deposited abstract retrieved through Crossref; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	Here using brain single-cell transcriptomes from five chordates-human 2 , mouse 3 , lizard 4 , lamprey 5 and amphioxus-we report that many cell-type families with conserved core transcription factors in vertebrates do not show one-to-one homology with amphioxus.
WHY IT MATTERS	The study advances mechanistic understanding of Whole-genome duplication shaped cell-type evolution in the vertebrate brain and creates a defined basis for testing conservation and developmental function in a vertebrate embryo.
MAIN / NEW METHODS	• Single-cell transcriptomics
KEY EVIDENCE	• Here using brain single-cell transcriptomes from five chordates-human 2 , mouse 3 , lizard 4 , lamprey 5 and amphioxus-we report that many cell-type families with conserved core transcription factors in vertebrates do not show one-to-one homology with amphioxus. • The findings indicate that ohnologues had a role in early vertebrate cell-type diversification. • Moreover, by examining paralogue expression across cell types and species, we show that expression changes were mainly driven by dosage selection and subfunctionalization.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Whole-genome duplication shaped cell-type evolution in the vertebrate brain

IMPORTANCE / SIGNIFICANCE

Neural Development And Function is central to how embryonic cells convert molecular state into reproducible tissue organization. Establishing whether the featured mechanism operates during early vertebrate development could reveal a conserved control point and a tractable route to explain developmental variability without overstating clinical translation.

KNOWLEDGE GAP

The featured paper does not establish whether the focal regulator or mechanism described by the featured paper acts during early vertebrate development, whether its effect is cell-autonomous, which developmental window is sensitive, or whether molecular and morphogenetic phenotypes can be rescued independently.

RATIONALE AND XENOPUS ADVANTAGE

The paper supplies a candidate mechanism, while early *Xenopus* provides rapid external development, targeted blastomere injection, explants, lineage tracing, live imaging, inexpensive biological replication across clutches, and direct comparison of molecular and tissue-scale outcomes.

CENTRAL HYPOTHESIS

Speculative: the focal regulator or mechanism described by the featured paper acts cell-autonomously during stages 10-18 to stabilize neural-domain geometry, lineage allocation, and reporter dynamics; acute loss will reduce the predicted molecular readout and increase pattern variance, while stage-matched wild-type rescue will restore both.

EXPERIMENTAL DESIGN

Use *Xenopus tropicalis* for F0 genetics and *Xenopus laevis* explants. Target neural plate and neural-border lineages by injecting two independent CRISPR RNPs at the 4- to 8-cell stage with a fluorescent lineage tracer; analyze stages 10-18. Groups: uninjected, non-targeting RNP, each guide alone, pooled guides, dose series, wild-type mRNA rescue, perturbation-resistant rescue, and temporally delayed perturbation where a validated small molecule or inducible construct exists. Primary readouts: neural-domain geometry, lineage allocation, and reporter dynamics. Secondary readouts: marker expression, cell trajectories, and neural-tube morphology, amplicon sequencing, RT-qPCR or in situ hybridization, and an orthogonal biochemical or reporter assay. Use at least three independent spawnings, 20-30 embryos per group per spawning for morphology, and 8-10 pooled or individually profiled samples per molecular condition; randomize embryos, blind image scoring, and analyze clutch as a random effect with multiplicity-controlled planned contrasts. Decision point: advance to epistasis and spatial single-cell profiling only if both guides reproduce the effect and rescue restores at least 50% of the molecular and developmental phenotype.

EXPECTED RESULTS AND INTERPRETATIONS

Support would be a guide-concordant loss of the predicted molecular readout, increased spatial or embryo-to-embryo variance, and rescue of both molecular and tissue phenotypes. This would place the mechanism upstream of developmental robustness. A molecular phenotype without morphogenesis would suggest buffering or a parallel tissue effector and would redirect the next aim to sensitized or explant conditions. A morphogenetic phenotype without the predicted reporter change would argue for an alternative biochemical function. No reproducible phenotype after confirmed editing would refute an obligate early role and shift the model toward redundancy, later function, or species-specific deployment.

POTENTIAL PITFALLS

- F0 mosaicism may weaken or spatially blur phenotypes.
- Paralog redundancy or maternal stores may mask an early requirement.

- High-dose perturbation may cause nonspecific toxicity or developmental delay.
- The featured mechanism may not be conserved or may use different effectors in amphibians.
- Abstract-only evidence may omit critical context needed to choose the most specific assay.

ALTERNATIVE STRATEGIES

- Map editing per embryo and enrich analysis for high-editing specimens; establish stable alleles if mosaicism remains limiting.
- Target expressed paralogs jointly, use maternal-zygotic strategies where feasible, or test dominant-negative and degron-based acute perturbations.
- Use lower doses, later induction, tissue-restricted injection, explants, and matched survival/staging controls to separate mechanism from toxicity.
- Test *Xenopus laevis* and *X. tropicalis* orthologs, then validate the key interaction biochemically or in a complementary cell model.
- Before full-scale work, manually review the linked paper's figures and methods and revise the reporter or perturbation to match the demonstrated mechanism.

PUBLISHED ENABLING EVIDENCE AND PROPOSED PRELIMINARY-DATA PLAN

Published enabling evidence and proposed preliminary-data plan: no unpublished user-laboratory data are assumed. The featured paper is the cited cross-species enabling source; direct *Xenopus* evidence was not automatically verified. Pilot 1: map candidate expression or pathway activity at 5-6 early stages using RT-qPCR and in situ hybridization (three spawnings; expected Figure 1A-B); go if a reproducible spatial or temporal signal exceeds twofold over the lowest stage. Pilot 2: test two CRISPR RNPs plus non-targeting control in three spawnings with at least 25 embryos per group, amplicon sequencing, blinded morphology, and the primary molecular readout (Figure 1C-E); go if both guides produce the same directional effect of at least 20% with acceptable survival. Pilot 3: perform wild-type mRNA rescue in three spawnings with at least 20 embryos per group (Figure 1F-G); go if rescue reverses at least 50% of the molecular and phenotype effect. Failure at any threshold triggers the paired alternative strategy rather than a claim of preliminary support.

NOVELTY medium-high - transfers a recent mechanism into a spatially and temporally tractable vertebrate embryo	FEASIBILITY high for the pilots - injection, explants, imaging, genotyping, and rescue are routine <i>Xenopus</i> capabilities	RISK medium - conservation, maternal contribution, and mosaicism are explicit decision points
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GRANT FIT

Exploratory Aim 1 for conservation and causality; successful pilots support Aim 2 epistasis and Aim 3 spatial mechanism.

STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-10629-x](https://doi.org/10.1038/s41586-026-10629-x)

A mosaic of whole-body representations on the human precentral gyrus

Darrel R. Deo, Elizaveta V. Okorokova, Anna L. Pritchard, Nick V. Hahn, Nicholas S. Card, Samuel R. Nason-Tomaszewski, Justin Jude, Thomas Hosman, et al.

journal-article | Nature 656(8128) | 2026-08-20

DOI 10.1038/s41586-026-10653-x

Access: Abstract retrieved from Europe PMC; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	Understanding how the body is represented in the motor cortex is key to understanding how the brain controls movement.
WHY IT MATTERS	The resulting map also provides important targeting information for brain-computer interfaces that seek to restore motor function.
MAIN / NEW METHODS	• Publisher-deposited title/abstract analysis; consult the linked paper for complete technical details
KEY EVIDENCE	• Understanding how the body is represented in the motor cortex is key to understanding how the brain controls movement. • We found that body parts were highly intermixed, such that the entire body was represented in all sampled locations of the PCG, although the relative strength of body parts was roughly consistent with the motor homunculus 17,18 . • We also found two speech-preferential areas with a broadly tuned, orofacial-dominant area in between them.
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GRANT FIT

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STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-10653-x](#)

Analysis of 173,303 exomes and genomes in the Pakistan Genome Resource

Christopher Koch, Shareef Khalid, Maleeha Zaman Khan, Shruthi Bandyadka, Brian Doyon, Daniel P. Denning, Muhammad Jahanzaib, Muhammad Rehan Mian, et al.

journal-article | Nature 656(8128) | 2026-08-20

DOI 10.1038/s41586-026-10667-5

Access: Abstract retrieved from Europe PMC; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	The Pakistan Genome Resource expands the catalogue of human genetic variants, provides a comprehensive genetic reference resource for the Pakistani population, and demonstrates the value of studying diverse cohorts to advance human health.
WHY IT MATTERS	The Pakistan Genome Resource expands the catalogue of human genetic variants, provides a comprehensive genetic reference resource for the Pakistani population, and demonstrates the value of studying diverse cohorts to advance human health.
MAIN / NEW METHODS	• Publisher-deposited title/abstract analysis; consult the linked paper for complete technical details
KEY EVIDENCE	• Naturally occurring loss-of-function variants in human genes enable drug target discovery because they mimic pharmacological inhibition of proteins. • Sequencing of diverse populations, particularly those enriched in familial relatedness, has been postulated to promote discovery of rare genetic variants 1-3 . • The Pakistan Genome Resource expands the catalogue of human genetic variants, provides a comprehensive genetic reference resource for the Pakistani population, and demonstrates the value of studying diverse cohorts to advance human health.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Analysis of 173,303 exomes and genomes in the Pakistan Genome Resource

IMPORTANCE / SIGNIFICANCE

Gene Regulation And Chromatin is central to how embryonic cells convert molecular state into reproducible tissue organization. Establishing whether the featured mechanism operates during early vertebrate development could reveal a conserved control point and a tractable route to explain developmental variability without overstating clinical translation.

KNOWLEDGE GAP

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RATIONALE AND XENOPUS ADVANTAGE

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CENTRAL HYPOTHESIS

Speculative: the focal regulator or mechanism described by the featured paper acts cell-autonomously during stages 8-14 to stabilize pathway activity, lineage allocation, and spatial marker-domain size; acute loss will reduce the predicted molecular readout and increase pattern variance, while stage-matched wild-type rescue will restore both.

EXPERIMENTAL DESIGN

Use *Xenopus tropicalis* for F0 genetics and *Xenopus laevis* explants. Target animal-cap, organizer, and mesendodermal lineages by injecting two independent CRISPR RNPs at the 4- to 8-cell stage with a fluorescent lineage tracer; analyze stages 8-14. Groups: uninjected, non-targeting RNP, each guide alone, pooled guides, dose series, wild-type mRNA rescue, perturbation-resistant rescue, and temporally delayed perturbation where a validated small molecule or inducible construct exists. Primary readouts: pathway activity, lineage allocation, and spatial marker-domain size. Secondary readouts: cell behavior, developmental timing, and gastrulation morphology, amplicon sequencing, RT-qPCR or in situ hybridization, and an orthogonal biochemical or reporter assay. Use at least three independent spawnings, 20-30 embryos per group per spawning for morphology, and 8-10 pooled or individually profiled samples per molecular condition; randomize embryos, blind image scoring, and analyze clutch as a random effect with multiplicity-controlled planned contrasts. Decision point: advance to epistasis and spatial single-cell profiling only if both guides reproduce the effect and rescue restores at least 50% of the molecular and developmental phenotype.

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NOVELTY medium-high - transfers a recent mechanism into a spatially and temporally tractable vertebrate embryo	FEASIBILITY high for the pilots - injection, explants, imaging, genotyping, and rescue are routine <i>Xenopus</i> capabilities	RISK medium - conservation, maternal contribution, and mosaicism are explicit decision points
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GRANT FIT

Exploratory Aim 1 for conservation and causality; successful pilots support Aim 2 epistasis and Aim 3 spatial mechanism.

STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-10667-5](https://doi.org/10.1038/s41586-026-10667-5)

Small-molecule modulation of beta-arrestins

Alem W. Kahsai, Natalia Pakharukova, Henry Y. Kwon, Kunal S. Shah, Caroline T. del Real, Bowie N. Shreiber, Jason G. Liang-Lin, Paul J. Shim, et al.

journal-article | Nature 656(8128) | 2026-08-20

DOI 10.1038/s41586-026-10683-5

Access: Publisher-deposited abstract retrieved through Crossref; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	Here we report the identification of small-molecule inhibitors that selectively target beta-arrestins and delineate their mechanism of action through integrated pharmacological, biochemical, biophysical and structural analyses.
WHY IT MATTERS	Together, these findings establish a mechanistic framework for beta-arrestin modulation, reveal a novel allosteric site for structure-based drug design, and open new avenues for transducer-targeted, pathway-specific GPCR therapeutic agents.
MAIN / NEW METHODS	• Electron microscopy • Biochemical reconstitution • Quantitative imaging • Computational modeling
KEY EVIDENCE	• Here we report the identification of small-molecule inhibitors that selectively target beta-arrestins and delineate their mechanism of action through integrated pharmacological, biochemical, biophysical and structural analyses. • Cryo-electron microscopy, molecular dynamics simulations and structure-guided mutagenesis reveal that one modulator, Cmpd-5, engages a pocket within the central crest of beta-arrestin1 formed by the middle, C and lariat loops, a critical receptor-binding interface, stabilizing a distinct conformation that is incompatible with full beta-arrestin-receptor engagement. • Together, these findings establish a mechanistic framework for beta-arrestin modulation, reveal a novel allosteric site for structure-based drug design, and open new avenues for transducer-targeted, pathway-specific GPCR therapeutic agents.
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Developmental timing and tissue autonomy of Small-molecule modulation of beta-arrestins

IMPORTANCE / SIGNIFICANCE

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Competing programs shape cortical sensorimotor-association axis development

Jeremiah Tsyporin, Menglei Zhang, Cai Qi, Ashlea Segal, Xinyun Li, Hyojin Kim, Sang-Hun Choi, Ivan Pavlovic, et al.

journal-article Nature 656(8128) 2026-08-20	
DOI 10.1038/s41586-026-10699-x	
Access: Abstract retrieved from Europe PMC; no full-text review is claimed. Evidence status: abstract-supported	
MAIN DISCOVERY	Here we present converging multispecies evidence supporting the multinodal induction-exclusion in network development (MIND) model, in which S-A patterning is governed by competing processes of induction and exclusion driven by two opposing transcriptomically defined programs. 'Pericentral' programs are induced around the frontotemporal poles, progress inwards toward the central regions of the undifferentiated neocortex and define higher-order association features. 'Central' programs are induced centrally through first-order sensorimotor thalamocortical inputs, establish primary areas and exclude pericentral programs.
WHY IT MATTERS	The MIND model provides a unifying framework for experimental, evolutionary and clinical phenomena, revealing induction and exclusion as antagonistic yet complementary principles shaping the S-A axis and processing hierarchies.
MAIN / NEW METHODS	• Publisher-deposited title/abstract analysis; consult the linked paper for complete technical details
KEY EVIDENCE	• Here we present converging multispecies evidence supporting the multinodal induction-exclusion in network development (MIND) model, in which S-A patterning is governed by competing processes of induction and exclusion driven by two opposing transcriptomically defined programs. 'Pericentral' programs are induced around the frontotemporal poles, progress inwards toward the central regions of the undifferentiated neocortex and define higher-order association features. 'Central' programs are induced centrally through first-order sensorimotor thalamocortical inputs, establish primary areas and exclude pericentral programs. • Induction and exclusion together establish an S-A organization in which primary areas emerge as focal islands within a broader ocean of distributed association networks. • The MIND model provides a unifying framework for experimental, evolutionary and clinical phenomena, revealing induction and exclusion as antagonistic yet complementary principles shaping the S-A axis and processing hierarchies.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Competing programs shape cortical sensorimotor-association axis development

IMPORTANCE / SIGNIFICANCE

Cell Signaling And Fate Control is central to how embryonic cells convert molecular state into reproducible tissue organization. Establishing whether the featured mechanism operates during early vertebrate development could reveal a conserved control point and a tractable route to explain developmental variability without overstating clinical translation.

KNOWLEDGE GAP

The featured paper does not establish whether the focal regulator or mechanism described by the featured paper acts during early vertebrate development, whether its effect is cell-autonomous, which developmental window is sensitive, or whether molecular and morphogenetic phenotypes can be rescued independently.

RATIONALE AND XENOPUS ADVANTAGE

The paper supplies a candidate mechanism, while early *Xenopus* provides rapid external development, targeted blastomere injection, explants, lineage tracing, live imaging, inexpensive biological replication across clutches, and direct comparison of molecular and tissue-scale outcomes.

CENTRAL HYPOTHESIS

Speculative: the focal regulator or mechanism described by the featured paper acts cell-autonomously during stages 8-14 to stabilize pathway activity, lineage allocation, and spatial marker-domain size; acute loss will reduce the predicted molecular readout and increase pattern variance, while stage-matched wild-type rescue will restore both.

EXPERIMENTAL DESIGN

Use *Xenopus tropicalis* for F0 genetics and *Xenopus laevis* explants. Target animal-cap, organizer, and mesendodermal lineages by injecting two independent CRISPR RNPs at the 4- to 8-cell stage with a fluorescent lineage tracer; analyze stages 8-14. Groups: uninjected, non-targeting RNP, each guide alone, pooled guides, dose series, wild-type mRNA rescue, perturbation-resistant rescue, and temporally delayed perturbation where a validated small molecule or inducible construct exists. Primary readouts: pathway activity, lineage allocation, and spatial marker-domain size. Secondary readouts: cell behavior, developmental timing, and gastrulation morphology, amplicon sequencing, RT-qPCR or in situ hybridization, and an orthogonal biochemical or reporter assay. Use at least three independent spawnings, 20-30 embryos per group per spawning for morphology, and 8-10 pooled or individually profiled samples per molecular condition; randomize embryos, blind image scoring, and analyze clutch as a random effect with multiplicity-controlled planned contrasts. Decision point: advance to epistasis and spatial single-cell profiling only if both guides reproduce the effect and rescue restores at least 50% of the molecular and developmental phenotype.

EXPECTED RESULTS AND INTERPRETATIONS

Support would be a guide-concordant loss of the predicted molecular readout, increased spatial or embryo-to-embryo variance, and rescue of both molecular and tissue phenotypes. This would place the mechanism upstream of developmental robustness. A molecular phenotype without morphogenesis would suggest buffering or a parallel tissue effector and would redirect the next aim to sensitized or explant conditions. A morphogenetic phenotype without the predicted reporter change would argue for an alternative biochemical function. No reproducible phenotype after confirmed editing would refute an obligate early role and shift the model toward redundancy, later function, or species-specific deployment.

POTENTIAL PITFALLS

- F0 mosaicism may weaken or spatially blur phenotypes.
- Paralog redundancy or maternal stores may mask an early requirement.

- High-dose perturbation may cause nonspecific toxicity or developmental delay.
- The featured mechanism may not be conserved or may use different effectors in amphibians.
- Abstract-only evidence may omit critical context needed to choose the most specific assay.

ALTERNATIVE STRATEGIES

- Map editing per embryo and enrich analysis for high-editing specimens; establish stable alleles if mosaicism remains limiting.
- Target expressed paralogs jointly, use maternal-zygotic strategies where feasible, or test dominant-negative and degron-based acute perturbations.
- Use lower doses, later induction, tissue-restricted injection, explants, and matched survival/staging controls to separate mechanism from toxicity.
- Test *Xenopus laevis* and *X. tropicalis* orthologs, then validate the key interaction biochemically or in a complementary cell model.
- Before full-scale work, manually review the linked paper's figures and methods and revise the reporter or perturbation to match the demonstrated mechanism.

PUBLISHED ENABLING EVIDENCE AND PROPOSED PRELIMINARY-DATA PLAN

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NOVELTY medium-high - transfers a recent mechanism into a spatially and temporally tractable vertebrate embryo	FEASIBILITY high for the pilots - injection, explants, imaging, genotyping, and rescue are routine <i>Xenopus</i> capabilities	RISK medium - conservation, maternal contribution, and mosaicism are explicit decision points
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GRANT FIT

Exploratory Aim 1 for conservation and causality; successful pilots support Aim 2 epistasis and Aim 3 spatial mechanism.

STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-10699-x](https://doi.org/10.1038/s41586-026-10699-x)

Chromatin landscape and epigenetic heterogeneity of acute myeloid leukaemia

Yotaro Ochi, Markus Liew-Littorin, Yasuhito Nannya, Sofia Bengtzen, Benedicte Piauger, Stefan Deneberg, Martin Jädersten, Vladimir Lazarevic, et al.

journal-article | Nature 656(8128) | 2026-08-20

DOI 10.1038/s41586-026-10703-4

Access: Abstract retrieved from Europe PMC; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	Gene mutations have been shown to have a large effect on pathogenesis, inter-tumour heterogeneity and clinical outcomes in AML 1-8 ; however, the role of epigenetic alterations in these respects has been investigated less extensively.
WHY IT MATTERS	In summary, ATAC-based chromatin profiling, combined with multiomics data, provides insights into AML pathogenesis beyond genomics and constitutes a valuable resource for AML research.
MAIN / NEW METHODS	Chromatin profiling
KEY EVIDENCE	- Gene mutations have been shown to have a large effect on pathogenesis, inter-tumour heterogeneity and clinical outcomes in AML 1-8 ; however, the role of epigenetic alterations in these respects has been investigated less extensively. - Here we use ATAC-seq (assay for transposase-accessible chromatin with sequencing) in a cohort of 1,563 individuals with a recent diagnosis of AML (the 'eCHROMA' cohort) to show that AML can be classified into 16 subgroups on the basis of chromatin accessibility profiles. - Multiomics analyses of gene mutations, the transcriptome, DNA methylation and histone marks show that these ATAC subgroups exhibit distinct driver mutations, differentiation states, gene expression, DNA methylation and super-enhancer profiles, and are also associated with clinical outcomes.
LIMITATIONS	- This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. - The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. - The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Chromatin landscape and epigenetic heterogeneity of acute myeloid leukaemia

IMPORTANCE / SIGNIFICANCE

Gene Regulation And Chromatin is central to how embryonic cells convert molecular state into reproducible tissue organization. Establishing whether the featured mechanism operates during early vertebrate development could reveal a conserved control point and a tractable route to explain developmental variability without overstating clinical translation.

KNOWLEDGE GAP

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RATIONALE AND XENOPUS ADVANTAGE

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EXPERIMENTAL DESIGN

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EXPECTED RESULTS AND INTERPRETATIONS

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POTENTIAL PITFALLS

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ALTERNATIVE STRATEGIES

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GRANT FIT

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STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-10703-4](https://doi.org/10.1038/s41586-026-10703-4)

Targeted enzyme discovery using metal-coordination mining

Ioannis Kipourou, Michelle C. Y. Chang

journal-article | Nature 656(8128) | 2026-08-20

DOI 10.1038/s41586-026-10716-z

Access: Abstract retrieved from Europe PMC; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	We apply this framework to the AlphaFold2 Protein Structure Database to identify new members of the Fe II /alpha-ketoglutarate-dependent halogenase family, which selectively functionalize unactivated C(sp 3)-H-bonds, a crucial transformation in the production of pharmaceuticals and other high-value compounds 3,4 .
WHY IT MATTERS	We apply this framework to the AlphaFold2 Protein Structure Database to identify new members of the Fe II /alpha-ketoglutarate-dependent halogenase family, which selectively functionalize unactivated C(sp 3)-H-bonds, a crucial transformation in the production of pharmaceuticals and other high-value compounds 3,4 .
MAIN / NEW METHODS	• Publisher-deposited title/abstract analysis; consult the linked paper for complete technical details
KEY EVIDENCE	• Central to these efforts is the discovery and functional annotation of novel enzymes, which is essential for elucidating the connection between genotype and phenotype and for developing biocatalysts for industrial applications. • However, accurately predicting enzymatic function remains a major challenge, and the discovery of new enzymes often relies on serendipity. • Here we present a metal-coordination-guided strategy that uses atomic-level mechanistic principles to mine protein structure databases for the targeted discovery of metalloenzymes.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Targeted enzyme discovery using metal-coordination mining

IMPORTANCE / SIGNIFICANCE

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KNOWLEDGE GAP

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RATIONALE AND XENOPUS ADVANTAGE

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EXPERIMENTAL DESIGN

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POTENTIAL PITFALLS

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NOVELTY medium-high - transfers a recent mechanism into a spatially and temporally tractable vertebrate embryo	FEASIBILITY high for the pilots - injection, explants, imaging, genotyping, and rescue are routine <i>Xenopus</i> capabilities	RISK medium - conservation, maternal contribution, and mosaicism are explicit decision points
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GRANT FIT

Exploratory Aim 1 for conservation and causality; successful pilots support Aim 2 epistasis and Aim 3 spatial mechanism.

STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-10716-z](https://doi.org/10.1038/s41586-026-10716-z)

Casdatifan shows durable response linked to HIF-2alpha biology in kidney cancer

Toni K. Choueiri, Jaime Merchán, Amita Patnaik, Alexandra Drakaki, Brian I. Rini, Sun Young Rha, Jae Lyun Lee, Moshe C. Ornstein, et al.

journal-article | Nature 656(8128) | 2026-08-20

DOI 10.1038/s41586-026-10718-x

Access: Publisher-deposited abstract retrieved through Crossref; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	Here we show that monotherapy with casdatifan-an orally bioavailable, potent and selective HIF-2alpha inhibitor 2 -produces meaningful, durable antitumour activity with manageable safety in individuals with refractory metastatic ccRCC.
WHY IT MATTERS	The study advances mechanistic understanding of Casdatifan shows durable response linked to HIF-2alpha biology in kidney cancer and creates a defined basis for testing conservation and developmental function in a vertebrate embryo.
MAIN / NEW METHODS	• Publisher-deposited title/abstract analysis; consult the linked paper for complete technical details
KEY EVIDENCE	• Here we show that monotherapy with casdatifan-an orally bioavailable, potent and selective HIF-2alpha inhibitor 2 -produces meaningful, durable antitumour activity with manageable safety in individuals with refractory metastatic ccRCC. • Overall, our findings show that casdatifan achieves meaningful, durable responses with manageable safety. • These data establish a link between on-target HIF-2alpha pathway modulation, tumour biology and clinical efficacy.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Casdatifan shows durable response linked to HIF-2alpha biology in kidney cancer

IMPORTANCE / SIGNIFICANCE

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KNOWLEDGE GAP

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CENTRAL HYPOTHESIS

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EXPERIMENTAL DESIGN

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STRONGEST REVIEWER-FACING PREMISE

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SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-10718-x](https://doi.org/10.1038/s41586-026-10718-x)

Virus reactivation in acute and long COVID-19

Cole Maguire, Jing Chen, Nadine Rouphael, Brinkley A. Morse, Annmarie Hoch, Harry Pickering, Hoang Van Phan, Abigail Glascock, et al.

journal-article | Nature 656(8128) | 2026-08-20

DOI 10.1038/s41586-026-10740-z

Access: Abstract retrieved from Europe PMC; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	Notably, SARS-CoV-2 infection has been shown to reactivate chronic viruses such as Epstein-Barr virus and cytomegalovirus, yet the full extent, temporal dynamics and immunological impact of viral reactivation in COVID-19 remain incompletely understood 2-7 .
WHY IT MATTERS	This study provides immune, transcriptomic and metabolomic signatures of viral reactivation that could inform future strategies to prognosticate and treat acute COVID-19 and long COVID.
MAIN / NEW METHODS	• Single-nucleus multi-omics
KEY EVIDENCE	• Notably, SARS-CoV-2 infection has been shown to reactivate chronic viruses such as Epstein-Barr virus and cytomegalovirus, yet the full extent, temporal dynamics and immunological impact of viral reactivation in COVID-19 remain incompletely understood 2-7 . • Here, leveraging multi-omic longitudinal data from 1,154 hospitalized patients with COVID-19 from the Immunophenotyping Assessment in a COVID-19 Cohort (IMPACC) study, we reveal significant reactivation of Herpesviridae and Anelloviridae during acute COVID-19, with distinct temporal dynamics for different viruses, and demonstrate that reactivation correlates with disease severity, host immune effects and clinical outcomes. • Although our results do not establish causation between virus reactivation and clinical outcomes, we highlight the prevalence of chronic viral reactivation during acute COVID-19 and long COVID.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Virus reactivation in acute and long COVID-19

IMPORTANCE / SIGNIFICANCE

Immunity And Host Response is central to how embryonic cells convert molecular state into reproducible tissue organization. Establishing whether the featured mechanism operates during early vertebrate development could reveal a conserved control point and a tractable route to explain developmental variability without overstating clinical translation.

KNOWLEDGE GAP

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The paper supplies a candidate mechanism, while early *Xenopus* provides rapid external development, targeted blastomere injection, explants, lineage tracing, live imaging, inexpensive biological replication across clutches, and direct comparison of molecular and tissue-scale outcomes.

CENTRAL HYPOTHESIS

Speculative: the focal regulator or mechanism described by the featured paper acts cell-autonomously during stages 8-18 to stabilize cell-stress signaling, epithelial integrity, and lineage-specific transcription; acute loss will reduce the predicted molecular readout and increase pattern variance, while stage-matched wild-type rescue will restore both.

EXPERIMENTAL DESIGN

Use *Xenopus tropicalis* for F0 genetics and *Xenopus laevis* explants. Target epidermal and mesendodermal lineages before mature immunity by injecting two independent CRISPR RNPs at the 4- to 8-cell stage with a fluorescent lineage tracer; analyze stages 8-18. Groups: uninjected, non-targeting RNP, each guide alone, pooled guides, dose series, wild-type mRNA rescue, perturbation-resistant rescue, and temporally delayed perturbation where a validated small molecule or inducible construct exists. Primary readouts: cell-stress signaling, epithelial integrity, and lineage-specific transcription. Secondary readouts: survival, barrier morphology, and developmental timing, amplicon sequencing, RT-qPCR or in situ hybridization, and an orthogonal biochemical or reporter assay. Use at least three independent spawnings, 20-30 embryos per group per spawning for morphology, and 8-10 pooled or individually profiled samples per molecular condition; randomize embryos, blind image scoring, and analyze clutch as a random effect with multiplicity-controlled planned contrasts. Decision point: advance to epistasis and spatial single-cell profiling only if both guides reproduce the effect and rescue restores at least 50% of the molecular and developmental phenotype.

EXPECTED RESULTS AND INTERPRETATIONS

Support would be a guide-concordant loss of the predicted molecular readout, increased spatial or embryo-to-embryo variance, and rescue of both molecular and tissue phenotypes. This would place the mechanism upstream of developmental robustness. A molecular phenotype without morphogenesis would suggest buffering or a parallel tissue effector and would redirect the next aim to sensitized or explant conditions. A morphogenetic phenotype without the predicted reporter change would argue for an alternative biochemical function. No reproducible phenotype after confirmed editing would refute an obligate early role and shift the model toward redundancy, later function, or species-specific deployment.

POTENTIAL PITFALLS

- F0 mosaicism may weaken or spatially blur phenotypes.
- Paralog redundancy or maternal stores may mask an early requirement.

- High-dose perturbation may cause nonspecific toxicity or developmental delay.
- The featured mechanism may not be conserved or may use different effectors in amphibians.
- Abstract-only evidence may omit critical context needed to choose the most specific assay.

ALTERNATIVE STRATEGIES

- Map editing per embryo and enrich analysis for high-editing specimens; establish stable alleles if mosaicism remains limiting.
- Target expressed paralogs jointly, use maternal-zygotic strategies where feasible, or test dominant-negative and degron-based acute perturbations.
- Use lower doses, later induction, tissue-restricted injection, explants, and matched survival/staging controls to separate mechanism from toxicity.
- Test *Xenopus laevis* and *X. tropicalis* orthologs, then validate the key interaction biochemically or in a complementary cell model.
- Before full-scale work, manually review the linked paper's figures and methods and revise the reporter or perturbation to match the demonstrated mechanism.

PUBLISHED ENABLING EVIDENCE AND PROPOSED PRELIMINARY-DATA PLAN

Published enabling evidence and proposed preliminary-data plan: no unpublished user-laboratory data are assumed. The featured paper is the cited cross-species enabling source; direct *Xenopus* evidence was not automatically verified. Pilot 1: map candidate expression or pathway activity at 5-6 early stages using RT-qPCR and in situ hybridization (three spawnings; expected Figure 1A-B); go if a reproducible spatial or temporal signal exceeds twofold over the lowest stage. Pilot 2: test two CRISPR RNPs plus non-targeting control in three spawnings with at least 25 embryos per group, amplicon sequencing, blinded morphology, and the primary molecular readout (Figure 1C-E); go if both guides produce the same directional effect of at least 20% with acceptable survival. Pilot 3: perform wild-type mRNA rescue in three spawnings with at least 20 embryos per group (Figure 1F-G); go if rescue reverses at least 50% of the molecular and phenotype effect. Failure at any threshold triggers the paired alternative strategy rather than a claim of preliminary support.

NOVELTY medium-high - transfers a recent mechanism into a spatially and temporally tractable vertebrate embryo	FEASIBILITY high for the pilots - injection, explants, imaging, genotyping, and rescue are routine <i>Xenopus</i> capabilities	RISK medium - conservation, maternal contribution, and mosaicism are explicit decision points
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GRANT FIT

Exploratory Aim 1 for conservation and causality; successful pilots support Aim 2 epistasis and Aim 3 spatial mechanism.

STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-10740-z](https://doi.org/10.1038/s41586-026-10740-z)

Reply to: Artefacts in single-cell mtDNA analyses misinform phylogenies

Chen Weng, Jonathan S. Weissman, Vijay G. Sankaran

journal-article | Nature 656(8128) | 2026-08-20

DOI 10.1038/s41586-026-10776-1

Access: Metadata and title only; no abstract or full text was retrieved. | Evidence status: metadata/title-supported

MAIN DISCOVERY	The title and publisher metadata identify a primary-research study of Reply to: Artefacts in single-cell mtDNA analyses misinform phylogenies; no abstract was retrieved, so no more specific discovery claim is made.
WHY IT MATTERS	The study advances mechanistic understanding of Reply to: Artefacts in single-cell mtDNA analyses misinform phylogenies and creates a defined basis for testing conservation and developmental function in a vertebrate embryo.
MAIN / NEW METHODS	• Publisher-deposited title/abstract analysis; consult the linked paper for complete technical details
KEY EVIDENCE	• The retrieved publisher-deposited metadata identifies this item as a journal article in the detected volume and issue. • The DOI provides the canonical path for manual verification of figures and full methods.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Reply to: Artefacts in single-cell mtDNA analyses misinform phylogenies

IMPORTANCE / SIGNIFICANCE

Developmental Conservation And Mechanism is central to how embryonic cells convert molecular state into reproducible tissue organization. Establishing whether the featured mechanism operates during early vertebrate development could reveal a conserved control point and a tractable route to explain developmental variability without overstating clinical translation.

KNOWLEDGE GAP

The featured paper does not establish whether the focal regulator or mechanism described by the featured paper acts during early vertebrate development, whether its effect is cell-autonomous, which developmental window is sensitive, or whether molecular and morphogenetic phenotypes can be rescued independently.

RATIONALE AND XENOPUS ADVANTAGE

The paper supplies a candidate mechanism, while early *Xenopus* provides rapid external development, targeted blastomere injection, explants, lineage tracing, live imaging, inexpensive biological replication across clutches, and direct comparison of molecular and tissue-scale outcomes.

CENTRAL HYPOTHESIS

Speculative: the focal regulator or mechanism described by the featured paper acts cell-autonomously during stages 8-14 to stabilize pathway activity, lineage allocation, and spatial marker-domain size; acute loss will reduce the predicted molecular readout and increase pattern variance, while stage-matched wild-type rescue will restore both.

EXPERIMENTAL DESIGN

Use *Xenopus tropicalis* for F0 genetics and *Xenopus laevis* explants. Target animal-cap, organizer, and mesendodermal lineages by injecting two independent CRISPR RNPs at the 4- to 8-cell stage with a fluorescent lineage tracer; analyze stages 8-14. Groups: uninjected, non-targeting RNP, each guide alone, pooled guides, dose series, wild-type mRNA rescue, perturbation-resistant rescue, and temporally delayed perturbation where a validated small molecule or inducible construct exists. Primary readouts: pathway activity, lineage allocation, and spatial marker-domain size. Secondary readouts: cell behavior, developmental timing, and gastrulation morphology, amplicon sequencing, RT-qPCR or in situ hybridization, and an orthogonal biochemical or reporter assay. Use at least three independent spawnings, 20-30 embryos per group per spawning for morphology, and 8-10 pooled or individually profiled samples per molecular condition; randomize embryos, blind image scoring, and analyze clutch as a random effect with multiplicity-controlled planned contrasts. Decision point: advance to epistasis and spatial single-cell profiling only if both guides reproduce the effect and rescue restores at least 50% of the molecular and developmental phenotype.

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PUBLISHED ENABLING EVIDENCE AND PROPOSED PRELIMINARY-DATA PLAN

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NOVELTY medium-high - transfers a recent mechanism into a spatially and temporally tractable vertebrate embryo	FEASIBILITY high for the pilots - injection, explants, imaging, genotyping, and rescue are routine <i>Xenopus</i> capabilities	RISK medium - conservation, maternal contribution, and mosaicism are explicit decision points
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GRANT FIT

Exploratory Aim 1 for conservation and causality; successful pilots support Aim 2 epistasis and Aim 3 spatial mechanism.

STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-10776-1](https://doi.org/10.1038/s41586-026-10776-1)

Artefacts in single-cell mtDNA analyses misinform phylogenies

Caleb A. Lareau, Michael Spencer Chapman, Livius Penter, Tal Nawy, Dana Pe’er, Leif S. Ludwig

journal-article | Nature 656(8128) | 2026-08-20

DOI 10.1038/s41586-026-10777-0

Access: Metadata and title only; no abstract or full text was retrieved. | Evidence status: metadata/title-supported

MAIN DISCOVERY	The title and publisher metadata identify a primary-research study of Artefacts in single-cell mtDNA analyses misinform phylogenies; no abstract was retrieved, so no more specific discovery claim is made.
WHY IT MATTERS	The study advances mechanistic understanding of Artefacts in single-cell mtDNA analyses misinform phylogenies and creates a defined basis for testing conservation and developmental function in a vertebrate embryo.
MAIN / NEW METHODS	• Publisher-deposited title/abstract analysis; consult the linked paper for complete technical details
KEY EVIDENCE	• The retrieved publisher-deposited metadata identifies this item as a journal article in the detected volume and issue. • The DOI provides the canonical path for manual verification of figures and full methods.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Artefacts in single-cell mtDNA analyses misinform phylogenies

IMPORTANCE / SIGNIFICANCE

Developmental Conservation And Mechanism is central to how embryonic cells convert molecular state into reproducible tissue organization. Establishing whether the featured mechanism operates during early vertebrate development could reveal a conserved control point and a tractable route to explain developmental variability without overstating clinical translation.

KNOWLEDGE GAP

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RATIONALE AND XENOPUS ADVANTAGE

The paper supplies a candidate mechanism, while early *Xenopus* provides rapid external development, targeted blastomere injection, explants, lineage tracing, live imaging, inexpensive biological replication across clutches, and direct comparison of molecular and tissue-scale outcomes.

CENTRAL HYPOTHESIS

Speculative: the focal regulator or mechanism described by the featured paper acts cell-autonomously during stages 8-14 to stabilize pathway activity, lineage allocation, and spatial marker-domain size; acute loss will reduce the predicted molecular readout and increase pattern variance, while stage-matched wild-type rescue will restore both.

EXPERIMENTAL DESIGN

Use *Xenopus tropicalis* for F0 genetics and *Xenopus laevis* explants. Target animal-cap, organizer, and mesendodermal lineages by injecting two independent CRISPR RNPs at the 4- to 8-cell stage with a fluorescent lineage tracer; analyze stages 8-14. Groups: uninjected, non-targeting RNP, each guide alone, pooled guides, dose series, wild-type mRNA rescue, perturbation-resistant rescue, and temporally delayed perturbation where a validated small molecule or inducible construct exists. Primary readouts: pathway activity, lineage allocation, and spatial marker-domain size. Secondary readouts: cell behavior, developmental timing, and gastrulation morphology, amplicon sequencing, RT-qPCR or in situ hybridization, and an orthogonal biochemical or reporter assay. Use at least three independent spawnings, 20-30 embryos per group per spawning for morphology, and 8-10 pooled or individually profiled samples per molecular condition; randomize embryos, blind image scoring, and analyze clutch as a random effect with multiplicity-controlled planned contrasts. Decision point: advance to epistasis and spatial single-cell profiling only if both guides reproduce the effect and rescue restores at least 50% of the molecular and developmental phenotype.

EXPECTED RESULTS AND INTERPRETATIONS

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POTENTIAL PITFALLS

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- Paralog redundancy or maternal stores may mask an early requirement.

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- The featured mechanism may not be conserved or may use different effectors in amphibians.
- Abstract-only evidence may omit critical context needed to choose the most specific assay.

ALTERNATIVE STRATEGIES

- Map editing per embryo and enrich analysis for high-editing specimens; establish stable alleles if mosaicism remains limiting.
- Target expressed paralogs jointly, use maternal-zygotic strategies where feasible, or test dominant-negative and degron-based acute perturbations.
- Use lower doses, later induction, tissue-restricted injection, explants, and matched survival/staging controls to separate mechanism from toxicity.
- Test *Xenopus laevis* and *X. tropicalis* orthologs, then validate the key interaction biochemically or in a complementary cell model.
- Before full-scale work, manually review the linked paper's figures and methods and revise the reporter or perturbation to match the demonstrated mechanism.

PUBLISHED ENABLING EVIDENCE AND PROPOSED PRELIMINARY-DATA PLAN

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NOVELTY medium-high - transfers a recent mechanism into a spatially and temporally tractable vertebrate embryo	FEASIBILITY high for the pilots - injection, explants, imaging, genotyping, and rescue are routine <i>Xenopus</i> capabilities	RISK medium - conservation, maternal contribution, and mosaicism are explicit decision points
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GRANT FIT

Exploratory Aim 1 for conservation and causality; successful pilots support Aim 2 epistasis and Aim 3 spatial mechanism.

STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-10777-0](https://doi.org/10.1038/s41586-026-10777-0)

Intestinal stem cells count self-renewal divisions to switch multipotency

Dong Tong, Anqi Li, Quanquan Jiang, Qili Yuan, Xiaozhao Liu, Ximiao He, Jiejunyi Liang, Yunyun Han, et al.

journal-article | Nature 656(8128) | 2026-08-20

DOI 10.1038/s41586-026-10814-y

Access: Abstract retrieved from Europe PMC; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	Here we show that ISCs intrinsically count self-renewal divisions through an epigenetic mechanism to control multipotency switching.
WHY IT MATTERS	Our work identifies a histone-modification-based division counter that programs developmental fidelity in stem cells, with implications for engineered tissue growth and differentiation disorder therapies.
MAIN / NEW METHODS	Organoid or stem-cell model
KEY EVIDENCE	- Here we show that ISCs intrinsically count self-renewal divisions through an epigenetic mechanism to control multipotency switching. - Crucially, EMC-derived transient Notch signalling establishes active marks in ISCs to initiate the count, designating each EMC production as the cycle's start point. - Our work identifies a histone-modification-based division counter that programs developmental fidelity in stem cells, with implications for engineered tissue growth and differentiation disorder therapies.
LIMITATIONS	- This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. - The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. - The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Intestinal stem cells count self-renewal divisions to switch multipotency

IMPORTANCE / SIGNIFICANCE

Cell Mechanics And Morphogenesis is central to how embryonic cells convert molecular state into reproducible tissue organization. Establishing whether the featured mechanism operates during early vertebrate development could reveal a conserved control point and a tractable route to explain developmental variability without overstating clinical translation.

KNOWLEDGE GAP

The featured paper does not establish whether the focal regulator or mechanism described by the featured paper acts during early vertebrate development, whether its effect is cell-autonomous, which developmental window is sensitive, or whether molecular and morphogenetic phenotypes can be rescued independently.

RATIONALE AND XENOPUS ADVANTAGE

The paper supplies a candidate mechanism, while early *Xenopus* provides rapid external development, targeted blastomere injection, explants, lineage tracing, live imaging, inexpensive biological replication across clutches, and direct comparison of molecular and tissue-scale outcomes.

CENTRAL HYPOTHESIS

Speculative: the focal regulator or mechanism described by the featured paper acts cell-autonomously during stages 10-13 to stabilize cell polarity, convergence-extension, tissue strain, and migration persistence; acute loss will reduce the predicted molecular readout and increase pattern variance, while stage-matched wild-type rescue will restore both.

EXPERIMENTAL DESIGN

Use *Xenopus tropicalis* for F0 genetics and *Xenopus laevis* explants. Target dorsal mesoderm and involuting marginal-zone explants by injecting two independent CRISPR RNPs at the 4- to 8-cell stage with a fluorescent lineage tracer; analyze stages 10-13. Groups: uninjected, non-targeting RNP, each guide alone, pooled guides, dose series, wild-type mRNA rescue, perturbation-resistant rescue, and temporally delayed perturbation where a validated small molecule or inducible construct exists. Primary readouts: cell polarity, convergence-extension, tissue strain, and migration persistence. Secondary readouts: fate markers, adhesion dynamics, and whole-embryo axis elongation, amplicon sequencing, RT-qPCR or in situ hybridization, and an orthogonal biochemical or reporter assay. Use at least three independent spawnings, 20-30 embryos per group per spawning for morphology, and 8-10 pooled or individually profiled samples per molecular condition; randomize embryos, blind image scoring, and analyze clutch as a random effect with multiplicity-controlled planned contrasts. Decision point: advance to epistasis and spatial single-cell profiling only if both guides reproduce the effect and rescue restores at least 50% of the molecular and developmental phenotype.

EXPECTED RESULTS AND INTERPRETATIONS

Support would be a guide-concordant loss of the predicted molecular readout, increased spatial or embryo-to-embryo variance, and rescue of both molecular and tissue phenotypes. This would place the mechanism upstream of developmental robustness. A molecular phenotype without morphogenesis would suggest buffering or a parallel tissue effector and would redirect the next aim to sensitized or explant conditions. A morphogenetic phenotype without the predicted reporter change would argue for an alternative biochemical function. No reproducible phenotype after confirmed editing would refute an obligate early role and shift the model toward redundancy, later function, or species-specific deployment.

POTENTIAL PITFALLS

- F0 mosaicism may weaken or spatially blur phenotypes.
- Paralog redundancy or maternal stores may mask an early requirement.

- High-dose perturbation may cause nonspecific toxicity or developmental delay.
- The featured mechanism may not be conserved or may use different effectors in amphibians.
- Abstract-only evidence may omit critical context needed to choose the most specific assay.

ALTERNATIVE STRATEGIES

- Map editing per embryo and enrich analysis for high-editing specimens; establish stable alleles if mosaicism remains limiting.
- Target expressed paralogs jointly, use maternal-zygotic strategies where feasible, or test dominant-negative and degron-based acute perturbations.
- Use lower doses, later induction, tissue-restricted injection, explants, and matched survival/staging controls to separate mechanism from toxicity.
- Test *Xenopus laevis* and *X. tropicalis* orthologs, then validate the key interaction biochemically or in a complementary cell model.
- Before full-scale work, manually review the linked paper's figures and methods and revise the reporter or perturbation to match the demonstrated mechanism.

PUBLISHED ENABLING EVIDENCE AND PROPOSED PRELIMINARY-DATA PLAN

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NOVELTY medium-high - transfers a recent mechanism into a spatially and temporally tractable vertebrate embryo	FEASIBILITY high for the pilots - injection, explants, imaging, genotyping, and rescue are routine <i>Xenopus</i> capabilities	RISK medium - conservation, maternal contribution, and mosaicism are explicit decision points
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GRANT FIT

Exploratory Aim 1 for conservation and causality; successful pilots support Aim 2 epistasis and Aim 3 spatial mechanism.

STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-10814-y](https://doi.org/10.1038/s41586-026-10814-y)

Vaccination elicits HIV broadly neutralizing antibodies in primates

Jon M. Steichen, Patrick J. Madden, Claudia T. Flynn, Swastik Phulera, Monolina Shil, Oleksandr Kalyuzhnyi, Alessia Liguori, Carolyne Kifude, et al.

journal-article | Nature 656(8128) | 2026-08-20

DOI 10.1038/s41586-026-10837-5

Access: Abstract retrieved from Europe PMC; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	Although the approach has shown promise in clinical 12-17 and preclinical 18-34 studies, it faces many immunological challenges and, to date, has not succeeded in generating bnAbs in humans or non-transgenic animals.
WHY IT MATTERS	The study advances mechanistic understanding of Vaccination elicits HIV broadly neutralizing antibodies in primates and creates a defined basis for testing conservation and developmental function in a vertebrate embryo.
MAIN / NEW METHODS	• Publisher-deposited title/abstract analysis; consult the linked paper for complete technical details
KEY EVIDENCE	• Although the approach has shown promise in clinical 12-17 and preclinical 18-34 studies, it faces many immunological challenges and, to date, has not succeeded in generating bnAbs in humans or non-transgenic animals. • Here we report testing of an adjuvanted protein germline-targeting vaccine in outbred non-human primates. • These results demonstrate proof of principle that germline-targeting vaccines can reproducibly elicit prespecified classes of bnAbs to prespecified epitopes under endogenous conditions, supporting further optimization of this approach for HIV vaccine development.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Vaccination elicits HIV broadly neutralizing antibodies in primates

IMPORTANCE / SIGNIFICANCE

Gene Regulation And Chromatin is central to how embryonic cells convert molecular state into reproducible tissue organization. Establishing whether the featured mechanism operates during early vertebrate development could reveal a conserved control point and a tractable route to explain developmental variability without overstating clinical translation.

KNOWLEDGE GAP

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- Before full-scale work, manually review the linked paper's figures and methods and revise the reporter or perturbation to match the demonstrated mechanism.

PUBLISHED ENABLING EVIDENCE AND PROPOSED PRELIMINARY-DATA PLAN

Published enabling evidence and proposed preliminary-data plan: no unpublished user-laboratory data are assumed. The featured paper is the cited cross-species enabling source; direct *Xenopus* evidence was not automatically verified. Pilot 1: map candidate expression or pathway activity at 5-6 early stages using RT-qPCR and in situ hybridization (three spawnings; expected Figure 1A-B); go if a reproducible spatial or temporal signal exceeds twofold over the lowest stage. Pilot 2: test two CRISPR RNPs plus non-targeting control in three spawnings with at least 25 embryos per group, amplicon sequencing, blinded morphology, and the primary molecular readout (Figure 1C-E); go if both guides produce the same directional effect of at least 20% with acceptable survival. Pilot 3: perform wild-type mRNA rescue in three spawnings with at least 20 embryos per group (Figure 1F-G); go if rescue reverses at least 50% of the molecular and phenotype effect. Failure at any threshold triggers the paired alternative strategy rather than a claim of preliminary support.

NOVELTY medium-high - transfers a recent mechanism into a spatially and temporally tractable vertebrate embryo	FEASIBILITY high for the pilots - injection, explants, imaging, genotyping, and rescue are routine <i>Xenopus</i> capabilities	RISK medium - conservation, maternal contribution, and mosaicism are explicit decision points
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GRANT FIT

Exploratory Aim 1 for conservation and causality; successful pilots support Aim 2 epistasis and Aim 3 spatial mechanism.

STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-10837-5](https://doi.org/10.1038/s41586-026-10837-5)

Enhanced B cell priming induces broadly neutralizing HIV-1 apex antibodies

Lorie Marchitto, Kshitij Wagh, Ryan S. Roark, Severin Coleon, Hui Li, Ashwin N. Skelly, Michael P. Hogarty, Rumi Habib, et al.

journal-article | Nature 656(8128) | 2026-08-20

DOI 10.1038/s41586-026-10838-4

Access: Abstract retrieved from Europe PMC; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	Using three different delivery platforms-persistently replicating simian human immunodeficiency viruses (SHIVs), protein nanoparticles and mRNA-we show bnAb priming as early as 4 weeks after infection or immunization, and neutralization breadth in the plasma by 12 weeks.
WHY IT MATTERS	These results expand the use of the rhesus model for HIV-1 vaccine design and provide a molecular blueprint for inducing V2 apex bnAbs in rhesus and humans.
MAIN / NEW METHODS	• Cryo-electron microscopy
KEY EVIDENCE	• Using three different delivery platforms-persistently replicating simian human immunodeficiency viruses (SHIVs), protein nanoparticles and mRNA-we show bnAb priming as early as 4 weeks after infection or immunization, and neutralization breadth in the plasma by 12 weeks. • Monoclonal bnAbs isolated from these animals were similarly broad and potent, with cryo-EM structures representing three distinct lineages revealing canonical needle-like HCDR3 binding.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Enhanced B cell priming induces broadly neutralizing HIV-1 apex antibodies

IMPORTANCE / SIGNIFICANCE

Immunity And Host Response is central to how embryonic cells convert molecular state into reproducible tissue organization. Establishing whether the featured mechanism operates during early vertebrate development could reveal a conserved control point and a tractable route to explain developmental variability without overstating clinical translation.

KNOWLEDGE GAP

The featured paper does not establish whether the focal regulator or mechanism described by the featured paper acts during early vertebrate development, whether its effect is cell-autonomous, which developmental window is sensitive, or whether molecular and morphogenetic phenotypes can be rescued independently.

RATIONALE AND XENOPUS ADVANTAGE

The paper supplies a candidate mechanism, while early *Xenopus* provides rapid external development, targeted blastomere injection, explants, lineage tracing, live imaging, inexpensive biological replication across clutches, and direct comparison of molecular and tissue-scale outcomes.

CENTRAL HYPOTHESIS

Speculative: the focal regulator or mechanism described by the featured paper acts cell-autonomously during stages 8-18 to stabilize cell-stress signaling, epithelial integrity, and lineage-specific transcription; acute loss will reduce the predicted molecular readout and increase pattern variance, while stage-matched wild-type rescue will restore both.

EXPERIMENTAL DESIGN

Use *Xenopus tropicalis* for F0 genetics and *Xenopus laevis* explants. Target epidermal and mesendodermal lineages before mature immunity by injecting two independent CRISPR RNPs at the 4- to 8-cell stage with a fluorescent lineage tracer; analyze stages 8-18. Groups: uninjected, non-targeting RNP, each guide alone, pooled guides, dose series, wild-type mRNA rescue, perturbation-resistant rescue, and temporally delayed perturbation where a validated small molecule or inducible construct exists. Primary readouts: cell-stress signaling, epithelial integrity, and lineage-specific transcription. Secondary readouts: survival, barrier morphology, and developmental timing, amplicon sequencing, RT-qPCR or in situ hybridization, and an orthogonal biochemical or reporter assay. Use at least three independent spawnings, 20-30 embryos per group per spawning for morphology, and 8-10 pooled or individually profiled samples per molecular condition; randomize embryos, blind image scoring, and analyze clutch as a random effect with multiplicity-controlled planned contrasts. Decision point: advance to epistasis and spatial single-cell profiling only if both guides reproduce the effect and rescue restores at least 50% of the molecular and developmental phenotype.

EXPECTED RESULTS AND INTERPRETATIONS

Support would be a guide-concordant loss of the predicted molecular readout, increased spatial or embryo-to-embryo variance, and rescue of both molecular and tissue phenotypes. This would place the mechanism upstream of developmental robustness. A molecular phenotype without morphogenesis would suggest buffering or a parallel tissue effector and would redirect the next aim to sensitized or explant conditions. A morphogenetic phenotype without the predicted reporter change would argue for an alternative biochemical function. No reproducible phenotype after confirmed editing would refute an obligate early role and shift the model toward redundancy, later function, or species-specific deployment.

POTENTIAL PITFALLS

- F0 mosaicism may weaken or spatially blur phenotypes.
- Paralog redundancy or maternal stores may mask an early requirement.

- High-dose perturbation may cause nonspecific toxicity or developmental delay.
- The featured mechanism may not be conserved or may use different effectors in amphibians.
- Abstract-only evidence may omit critical context needed to choose the most specific assay.

ALTERNATIVE STRATEGIES

- Map editing per embryo and enrich analysis for high-editing specimens; establish stable alleles if mosaicism remains limiting.
- Target expressed paralogs jointly, use maternal-zygotic strategies where feasible, or test dominant-negative and degron-based acute perturbations.
- Use lower doses, later induction, tissue-restricted injection, explants, and matched survival/staging controls to separate mechanism from toxicity.
- Test *Xenopus laevis* and *X. tropicalis* orthologs, then validate the key interaction biochemically or in a complementary cell model.
- Before full-scale work, manually review the linked paper's figures and methods and revise the reporter or perturbation to match the demonstrated mechanism.

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CROSS-PAPER SYNTHESIS

RECURRING MECHANISMS

- gene regulation and chromatin (5 papers)
- neural development and function (2 papers)
- cell signaling and fate control (2 papers)
- immunity and host response (2 papers)
- developmental conservation and mechanism (2 papers)

METHODOLOGICAL TRENDS

- Publisher-deposited title/abstract analysis; consult the linked paper for complete technical details (8)
- Single-cell transcriptomics (1)
- Electron microscopy (1)
- Biochemical reconstitution (1)
- Quantitative imaging (1)
- Computational modeling (1)

RANKED XENOPUS OPPORTUNITIES

#1 Developmental timing and tissue autonomy of Whole-genome duplication shaped cell-type evolution in the vertebrate brain

Significance: high if the focal mechanism is conserved | Novelty: medium-high | Mechanistic Clarity: medium pending candidate-specific assay refinement | Preliminary Support: featured-paper evidence; Xenopus pilots proposed | Xenopus Advantage: high for temporal, spatial, rescue, and explant experiments | Feasibility: high for the three rapid pilots | Follow On Aims: epistasis, spatial profiling, and cross-species validation

#2 Developmental timing and tissue autonomy of A mosaic of whole-body representations on the human precentral gyrus

Significance: high if the focal mechanism is conserved | Novelty: medium-high | Mechanistic Clarity: medium pending candidate-specific assay refinement | Preliminary Support: featured-paper evidence; Xenopus pilots proposed | Xenopus Advantage: high for temporal, spatial, rescue, and explant experiments | Feasibility: high for the three rapid pilots | Follow On Aims: epistasis, spatial profiling, and cross-species validation

#3 Developmental timing and tissue autonomy of Analysis of 173,303 exomes and genomes in the Pakistan Genome Resource

Significance: high if the focal mechanism is conserved | Novelty: medium-high | Mechanistic Clarity: medium pending candidate-specific assay refinement | Preliminary Support: featured-paper evidence; Xenopus pilots proposed | Xenopus Advantage: high for temporal, spatial, rescue, and explant experiments | Feasibility: high for the three rapid pilots | Follow On Aims: epistasis, spatial profiling, and cross-species validation

#4 Developmental timing and tissue autonomy of Small-molecule modulation of beta-arrestins

Significance: high if the focal mechanism is conserved | Novelty: medium-high | Mechanistic Clarity: medium pending candidate-specific assay refinement | Preliminary Support: featured-paper evidence; Xenopus pilots proposed | Xenopus Advantage: high for temporal, spatial, rescue, and explant experiments | Feasibility: high for the three rapid pilots | Follow On Aims: epistasis, spatial profiling, and cross-species validation

COHERENT MULTI-AIM PROPOSAL

A coherent multi-aim proposal can unite papers centered on gene regulation and chromatin: Aim 1 establishes conservation and timing, Aim 2 resolves cell autonomy and pathway order, and Aim 3 links molecular activity to tissue-scale robustness.

INDEPENDENT PILOTS

- Developmental timing and tissue autonomy of A mosaic of whole-body representations on the human precentral gyrus
- Developmental timing and tissue autonomy of Analysis of 173,303 exomes and genomes in the Pakistan Genome Resource
- Developmental timing and tissue autonomy of Small-molecule modulation of beta-arrestins

SOURCE / METHOD NOTE

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